Part A

Weakness 1:

The paper's central logical-consistency metric, contradiction rate, is built on incorrectly constructed negation examples. In the code, each BoolQ input is created as question [SEP] passage, but in create_contradiction_pairs the negated input is built with prefix = text_fwd.split("[SEP]")[0] and text_neg = prefix + "[SEP] " + q_neg. That keeps only the original question and discards the passage, so the negated example becomes original question [SEP] negated question instead of negated question [SEP] original passage. As a result, the reported contradiction-rate numbers in both the DistilBERT and BERT/RoBERTa experiments are not valid measurements of logical consistency on the same evidence.

Because the paper's main claim, improved logical consistency, is supported by a broken evaluation setup, the headline result is not trustworthy.

Rewrite contradiction-pair construction so the passage is preserved and only the question is negated, then rerun all contradiction-rate experiments and include a small manually checked table of forward/negated pairs to verify the benchmark is constructed correctly.

Weakness 2:

The paper overclaims that MITR improves logical reasoning across models, but the evidence is both incomplete and contradictory. In the DistilBERT results, InfoNCE is the best method, with the top final accuracy and the largest contradiction-rate reduction, yet the BERT/RoBERTa notebook explicitly drops InfoNCE and tests only Cosine and CKA. At the same time, the 12-layer results reverse sign across backbones: on BERT, both MITR-Cosine and MITR-CKA worsen contradiction rate relative to baseline, while on RoBERTa both improve it. So the paper's cross-model story is built without the strongest method and the methods that are tested do not behave consistently across architectures.

Because the paper claims a general mechanism for improving reasoning, but the scaling evidence is incomplete, and the observed effect is not robust across backbones.

Run the full strategy set on BERT and RoBERTa, especially InfoNCE, then add a backbone-by-strategy ablation with the same training budget and a short analysis of why MITR helps RoBERTa but hurts BERT, or narrow the claim to the specific settings where the method reliably works.

Part B

ICLR 2026 Workshop on Logical Reasoning of Large Language Models

This paper is a strong fit for the ICLR 2026 Workshop on Logical Reasoning of Large Language Models because it targets exactly the failure mode the workshop highlights: self-contradictory behavior across related questions. The workshop explicitly calls for work on improving deduction, induction, and abduction in language models, avoiding logical contradictions across multiple related responses, and benchmarking logical reasoning and consistency. MITR is a training-time intervention aimed at reducing inter-layer redundancy so that transformer layers encode more distinct reasoning steps, and the paper evaluates that idea using both task accuracy and a contradiction-rate metric on BoolQ-style yes/no reasoning. Even though the experiments use BERT-family encoders rather than frontier decoder LLMs, the core scientific question is the same: can architectural or training changes make language models more logically consistent.

Logical Consistency of Large Language Models - This is directly related because it studies whether language models remain consistent under negation and other logical transformations, which closely matches MITR's goal of reducing contradictions across related questions.

Large Language Models Meet Symbolic Provers for Logical Reasoning Evaluation - This paper is closely related because it builds a rigorous symbolic benchmark for evaluating logical reasoning beyond surface accuracy, which aligns with MITR's goal of measuring and improving logical consistency rather than just end-task performance.

Part C

Experiment 1 -

Scale the strongest MITR variant (InfoNCE) to BERT and RoBERTa

If MITR-InfoNCE is the genuinely strongest redundancy-reduction strategy, then after correcting the contradiction benchmark it will outperform Cosine and CKA on BERT and RoBERTa as well, because the best method on DistilBERT should remain competitive when scaled to larger backbones.

I would expect InfoNCE to be at least one of the top-performing strategies on BERT and RoBERTa, and possibly the strongest overall if the DistilBERT result reflects a real mechanism rather than a small-model artifact. This experiment directly closes the biggest gap in the current submission: the best DistilBERT method was excluded from the 12-layer study, so the paper currently cannot support a cross-backbone claim about MITR.

If InfoNCE fails on BERT and RoBERTa, then MITR's benefits are not robust across model scales or architectures, and the paper's claim must be narrowed to a model-specific effect rather than a general transformer reasoning improvement.

2-3 hours on A100

Experiment 2 –

Direct redundancy validation: measure whether MITR reduces adjacent-layer similarity, and whether that reduction tracks logical consistency gains.

If MITR improves logical consistency by forcing adjacent transformer layers to encode more distinct information, then MITR-trained models should show lower adjacent-layer similarity than the baseline, and the models with the largest similarity reduction should also show the largest contradiction-rate improvement, because the paper's claimed mechanism is reduced inter-layer redundancy.

After training, I would extract hidden states from each layer on the validation set and compute similarity between adjacent layers using metrics such as cosine similarity and linear CKA. I would expect successful MITR settings, especially DistilBERT-InfoNCE and any successful BERT/RoBERTa variants, to show lower adjacent-layer similarity than the corresponding baseline models. I would also expect the size of this reduction to correlate with contradiction-rate improvement. This experiment would strengthen the paper by testing the proposed mechanism directly, rather than only reporting downstream accuracy and contradiction-rate changes.

1-2 hours on A100

Part D

This experiment tested the missing InfoNCE condition on the two 12-layer backbones omitted from the original scaling study. In this scoped 3-epoch Colab setting, MITR-InfoNCE improved BERT on both metrics, raising validation accuracy from 0.634 to 0.664 and reducing contradiction rate from 0.945 to 0.885. However, the same method did not transfer to RoBERTa, where accuracy fell slightly from 0.630 to 0.624 and contradiction rate worsened from 0.985 to 1.000. These results partially support the hypothesis: InfoNCE can help on larger backbones, but the effect is not robust across architectures. The experiment therefore strengthens the paper by closing its clearest missing condition while also sharpening the conclusion. Rather than showing that MITR-InfoNCE is a universally effective reasoning intervention, the results suggest that its gains are backbone-dependent, with BERT benefiting in this setup and RoBERTa not.

Part E

Logical inconsistency across related questions remains a major limitation of transformer language models, even when they perform reasonably well on binary reasoning tasks. We propose Mutual Information Transformer Regularization (MITR), a training-time method that penalizes redundancy between adjacent transformer layers so that successive layers encode more distinct information. Our central finding is that MITR can improve logical consistency, but its benefits are backbone-dependent rather than universal. Across BoolQ experiments on DistilBERT, BERT-base, and RoBERTa-base, the strongest result appears on DistilBERT, where MITR-InfoNCE improves contradiction rate from 0.4300 to 0.4120 while slightly increasing validation accuracy from 0.6980 to 0.7000. However, the 12-layer results are mixed: in our Part D extension, adding InfoNCE to BERT improves accuracy from 0.634 to 0.664 and lowers contradiction rate from 0.945 to 0.885, whereas RoBERTa does not benefit. Taken together, these results suggest that reducing inter-layer redundancy is a promising direction for improving reasoning consistency, but that its success depends strongly on the underlying model architecture.